Daniel Zhang

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Education

Bachelor of Engineering, Mechatronics Engineering CO-OP — Hamilton, ON
McMaster University, Level III — Expected Apr 2028

- Notable Coursework: Analog & Digital Circuits, Statics, Dynamics, Electricity and Magnetism, Thermodynamics, Embedded Systems, Engineering Mathematics, Data Structures & Algorithms

Skills

Industry & Safety: Project Engineering, Process Flow Automation, Technical Writing, WHMIS, Lockout/Tagout

Design, Modeling & Hardware: CAD (SOLIDWORKS — CSWA Certified, Inventor, Fusion 360), FEA, NI Multisim, Embedded Systems, STM32, Soldering, 3D Printing, GD&T

Software: Python, C/C++, MATLAB/Simulink, Julia, SMath, Maple, FlexPDE, Microsoft 365, Power Automate

Experience & Involvement

Mechanical Sub-Team Member — Wheel Assembly — Hamilton, ON
McMaster Chem-E-Car — Sep 2026 - Present

- Designing and modeling a custom wheel assembly in SolidWorks for a $40 \times 30 \times 18$ cm vehicle, carrying up to a 500 mL water payload for the AIChE National Competition.
- Designing an independent double wishbone suspension to isolate dynamic weight transfer, stabilizing the vehicle chassis to prevent false triggers on optical stopping sensors.
- Collaborated with a team of 3 to implement an adjustable manual transmission for rapid ratio adjustments, enabling on-site drivetrain tuning to target distances disclosed on competition day at the Chem-E-Car National Championship.

Nuclear Engineering Co-op Student — Cambridge, ON
Eclipse Automation — Jan 2026 - Present

- Authored and developed high-consequence engineering deliverables (FMEAs, FAT Plans, System Design Descriptions) for CSA N299.2 and N299.3 rated tools, driving successful milestones for the Pickering, Cernavodă, and Qinshan nuclear reactor refurbishment projects.
- Executed Finite Element Analysis (FEA) and Allowable Strength Design (ASD) calculations for specialized nuclear tooling on the Pickering (RFBR) refurbishment project, validating structural integrity and safety margins to ensure operational reliability.
- Delivered project engineering support for Design Verification Testing (DVT) of 4 specialized manufacturing tools, partnering with toolmakers, automation technicians, electricians, and clients to validate system performance against design requirements and facilitate customer sign-off.
- Facilitated CNSC regulatory approval for CNL's $100M SBWRS project by verifying 5,000+ technical requirements, and engineered an automated workflow to audit a library of 1,000+ engineering standards and flag discrepancies.

Tutor — Richmond Hill, ON
Academy for Mathematics & English — Jul 2024 - Dec 2025

- Provided one-on-one tutoring in mathematics and physics for elementary, high school and university-prep students, simplifying complex technical concepts into clear, practical explanations.
- Designed personalized lesson plans and study materials to target individual learning gaps, improving students' test scores and conceptual understanding over multiple sessions.

Projects

12-DOF Quadruped Robot — Aug 2026

- Designed a 12-DOF quadruped in Autodesk Fusion with a print-ready chassis, integrating MG90S servos, a PCA9685 driver, and a Raspberry Pi Zero 2 W.
- Developed a custom Inverse Kinematic (IK) solver for stable omnidirectional gait patterns, tested in MuJoCo, with OpenCV camera integration for autonomous obstacle avoidance.

Computer Vision-Controlled Robotic Arm — Apr 2025

- Prototyped a 2-DOF robotic arm with a team of 3 in a 12-hour hackathon, pairing a Python OpenCV vision pipeline with an ESP32 to track hand movements in real time.
- Engineered embedded C++ firmware to process coordinate packets over UDP and generate step pulses, tuning motion profiles to eliminate jitter and ensure smooth multi-axis tracking.